SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
Program Name: B. Tech	**Assignment Type: Lab**	**Academic Year:**2025-2026

Q.No.	Question	*Expected Time to complete*
1	**Lab 19 – Code Translation: Converting Between Programming Languages** **Lab Objectives:** • Understand how AI tools can assist in translating code between different programming languages. • Learn to verify correctness and functionality after translation. • Explore syntactic and semantic differences between languages (e.g., Python, Java, C++). • Practice debugging and optimizing AI-translated code. --- **Task 1 – Python to Java Conversion** **Task:** Translate a simple **Python class** into **Java** using AI assistance. **Instructions:** • Prompt AI to convert a Python class with attributes and methods into equivalent Java code. • Verify that constructors, method signatures, and data types are properly handled. • Run both versions (Python & Java) to compare expected outputs. **Expected Output:** • Equivalent working Java class translated from Python. --- **Task 2 – C++ to Python Function Conversion** **Task:** Convert a **C++ function** for factorial calculation into Python. **Instructions:** • Ask AI to translate a recursive C++ factorial function into Python. • Validate correctness by testing with sample inputs. • Ensure Python version follows Pythonic conventions.	Week10 - Monday

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

Task 3 – SQL to Pandas Query

Task:

Translate a **SQL query** into a **Pandas DataFrame operation** in Python.

Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

Task 4 – Real-Time Application: Multi-Language Snippet Converter

Scenario:

Students will choose a **real-time use case** where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into **JavaScript** for browser automation.
- Translating a **Java method** into **C#** for enterprise applications.

Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

- Working equivalent code snippets in at least **two different programming languages**.